Name: Balaji Patil Section: 10 Course: OSSP

ID number: 2520080065 Week 2

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

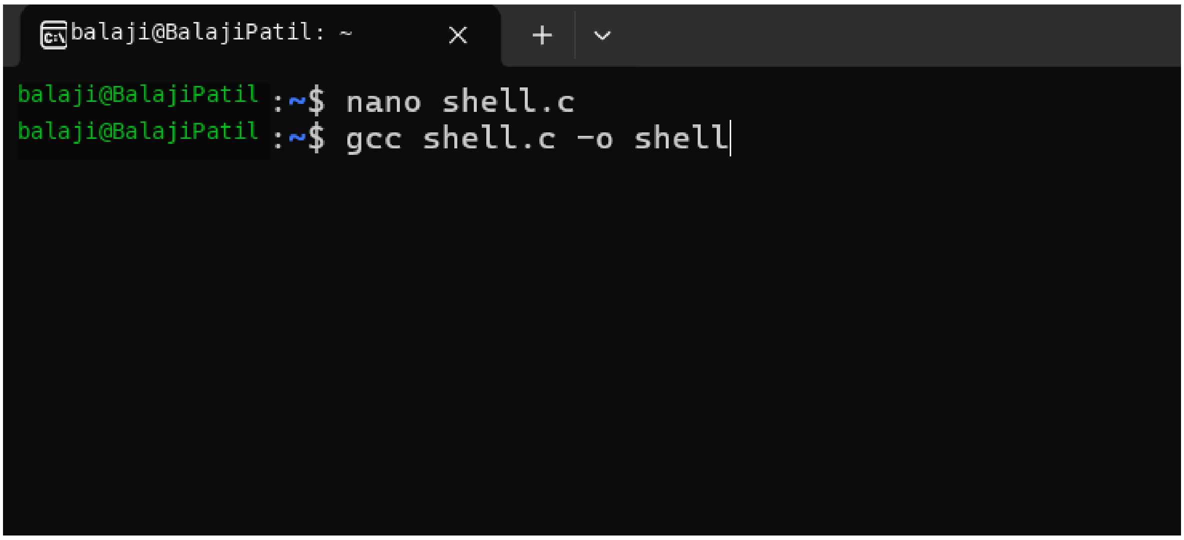
A) Create a file by using

```
balaji@BalajiPatil:~$ nano shell.c
```

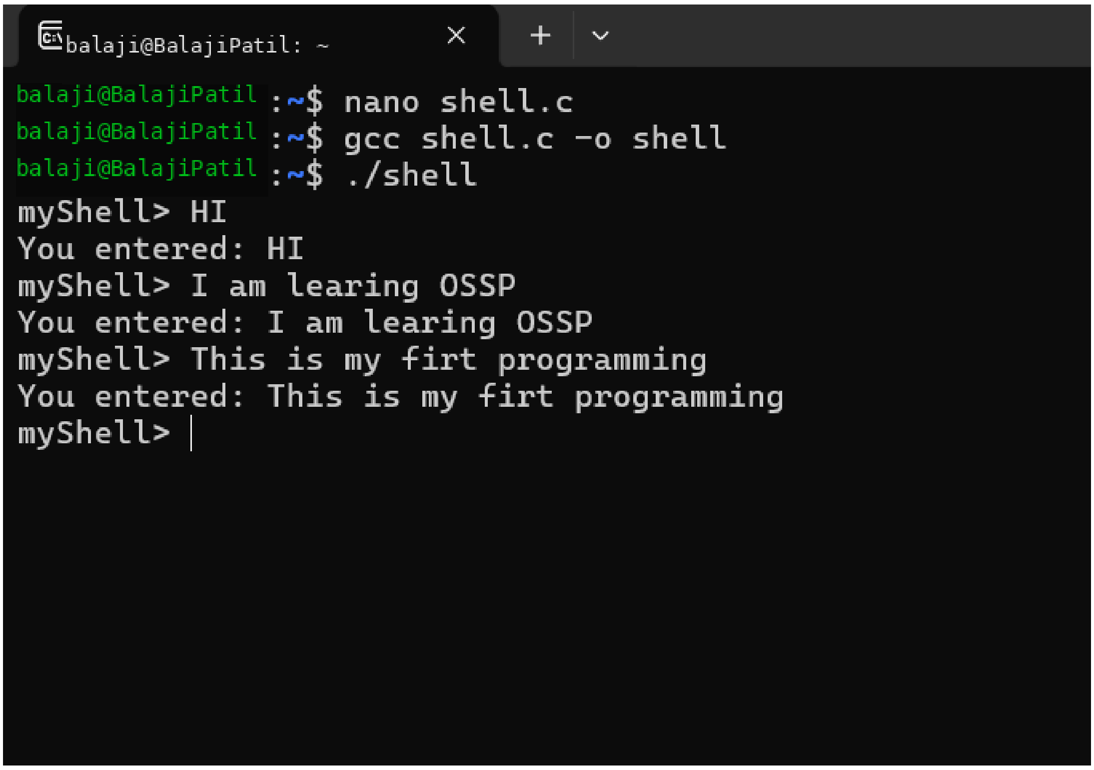
Then press enter and the code terminal will open then write a code

```
balaji@BalajiPatil: ~
GNU nano 7.2                                                    shel
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    while (1) {
        printf("myShell> ");
        fgets(input, sizeof(input), stdin);

        // Remove newline
        input[strcspn(input, "\n")] = 0;

        // Exit condition
        if (strcmp(input, "exit") == 0) {
            printf("Exiting Shell...\n");
            break;
        }

        printf("You entered: %s\n", input);
    }

    return 0;
}
```

After writing the code save the your program by ctrl O and for coming back to command press ctrl X then compile it by the command



After compiling to get the output enter command



Task 2: Using the following system calls, copy the content from File1.txt to File2.txt.

1. **Open()**
2. **Read()**
3. **Write()**
4. **Close()**

open()

syntax: open(“ filename.txt”, modes, permissons)

Example: open(“First_Prgm.txt”, O_CREAT | O_WRONLY, 0644);

Table 1: Modes of Open system call

Flag	Purpose
O_RDONLY	Open for read only
O_WRONLY	Open for write only
O_RDWR	Open for both read and write
O_CREAT	Create file if it does not exist
O_TRUNC	Truncate (empty) an existing file
O_APPEND	Append data to the end of the file
O_EXCL	Fail if file already exists (used with O_CREAT)

read()

Syntax: read(int fd, void *buf, size_t count);

Example: read(fd, buffer, 100)

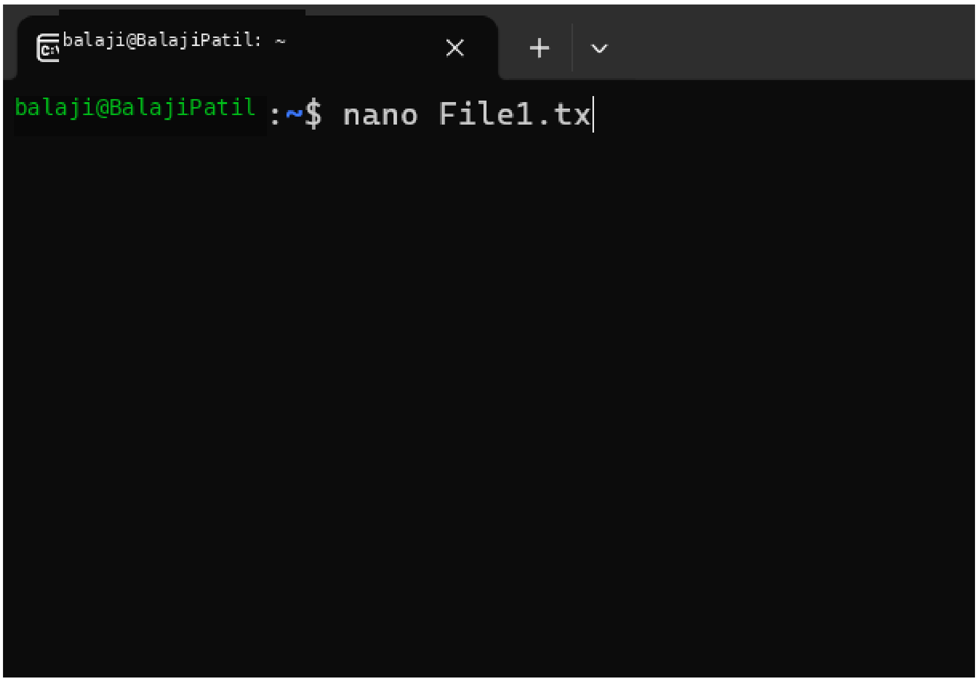
```
#include <stdio.h>
#include <unistd.h>
int main()
{
  char buffer[100];
  int n;
  printf("Enter some text: ");
  n = read(0, buffer, sizeof(buffer) - 1);
  buffer[n] = '\0';   // Null-terminate the string
  printf("You entered: %s", buffer);
  return 0;
}
```

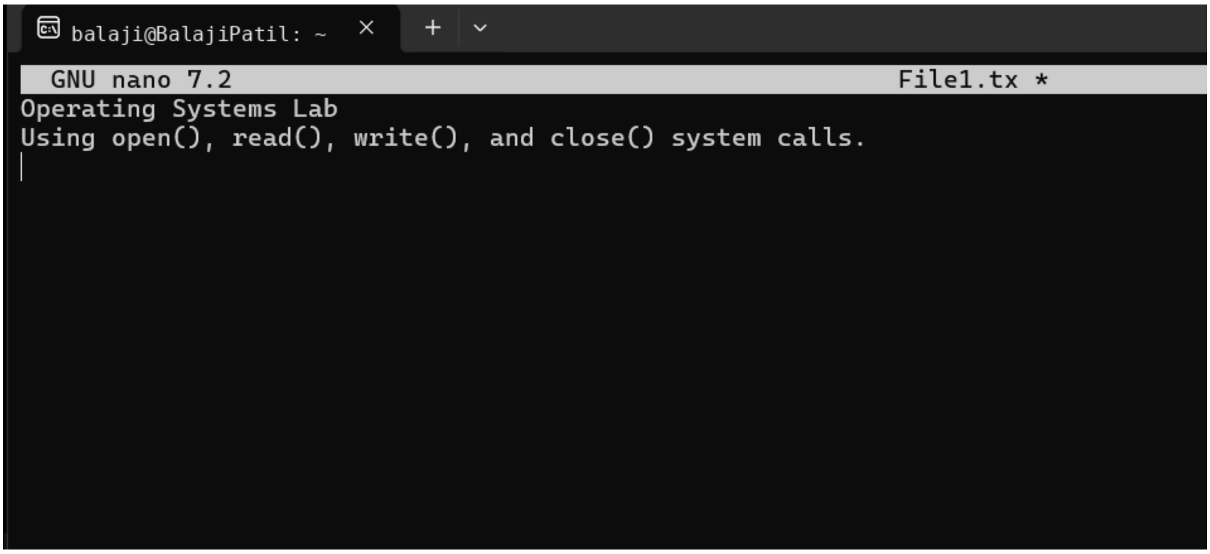
write()

Syntax: write(int fd, const void *buf, size_t count);

Example: write(fd, message, 11);

```
#include <unistd.h>
int main()
{
  write(1, "Hello World\n", 12);
  return 0;
}
```

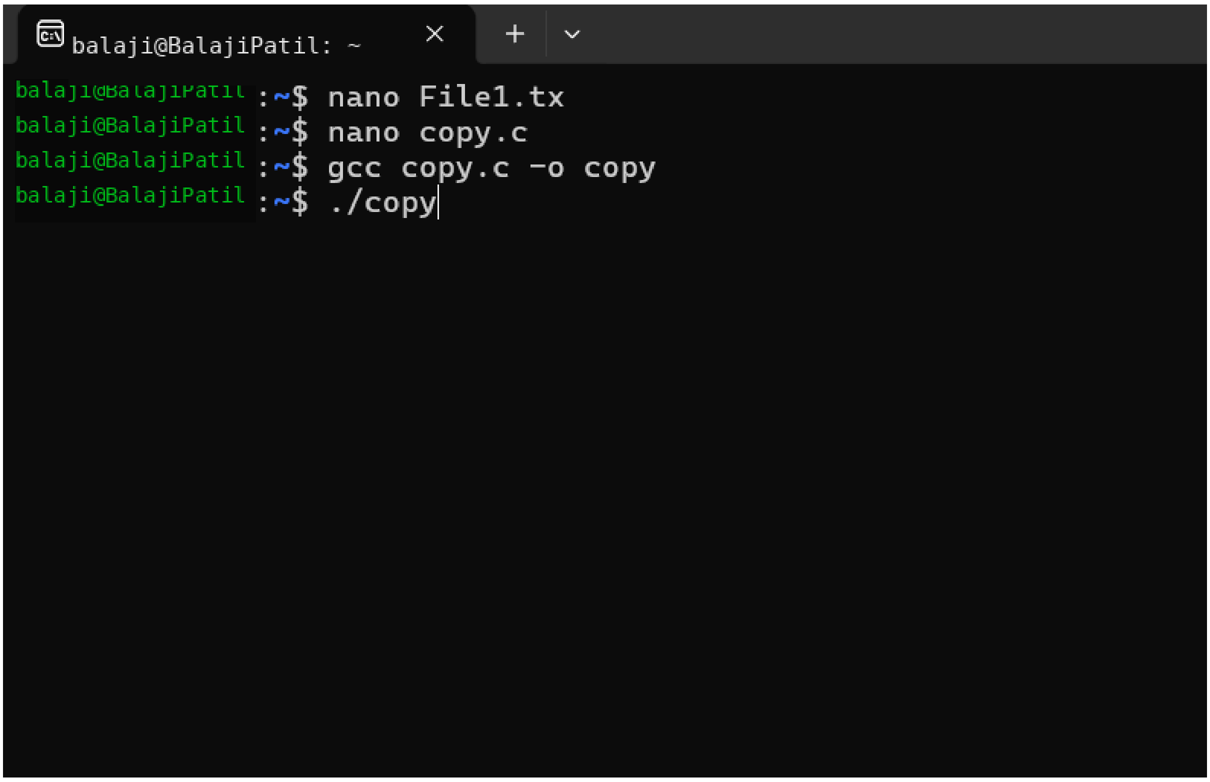

balaji@BalajiPatil: ~
balaji@BalajiPatil:~$ nano File1.tx


balaji@BalajiPatil: ~
GNU nano 7.2
File1.tx *
Operating Systems Lab
Using open(), read(), write(), and close() system calls.

```
balaji@BalajiPatil: ~
balaji@BalajiPatil:~$ nano File1.tx
balaji@BalajiPatil:~$ nano copy.c
balaji@BalajiPatil:~$
```

```
shi balaji@BalajiPatil: ~
  GNU nano 7.2                                        copy.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead;

    // Open File1.txt for reading
    source = open("File1.txt", O_RDONLY);

    if (source < 0)
    {
        printf("Error opening File1.txt\n");
        return 1;
    }

    // Open File2.txt for writing (create if it doesn't exist)
    destination = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);

    if (destination < 0)
    {
        printf("Error opening File2.txt\n");
                                                   [ Read 44 lines ]
^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo       M-A
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo       M-6
```


balaji@BalajiPatil: ~
balaji@BalajiPatil:~$ nano File1.tx
balaji@BalajiPatil:~$ nano copy.c
balaji@BalajiPatil:~$ gcc copy.c -o copy


balaji@BalajiPatil: ~
balaji@BalajiPatil:~$ nano File1.tx
balaji@BalajiPatil:~$ nano copy.c
balaji@BalajiPatil:~$ gcc copy.c -o copy
balaji@BalajiPatil:~$ ./copy

```
balaji@BalajiPatil: ~

balaji@BalajiPatil:~$ nano File1.tx
balaji@BalajiPatil:~$ nano copy.c
balaji@BalajiPatil:~$ gcc copy.c -o copy
balaji@BalajiPatil:~$ ./copy
Content copied successfully.
balaji@BalajiPatil:~$ cat File1.txt
```

```
balaji@BalajiPatil: ~

balaji@BalajiPatil:~$ nano File1.tx
balaji@BalajiPatil:~$ nano copy.c
balaji@BalajiPatil:~$ gcc copy.c -o copy
balaji@BalajiPatil:~$ ./copy
Content copied successfully.
balaji@BalajiPatil:~$ cat File1.txt
Operating Systems Lab
Using open(), read(), write(), and close() system calls.
balaji@BalajiPatil:~$ cat File2.txt
Operating Systems Lab
Using open(), read(), write(), and close() system calls.
balaji@BalajiPatil:~$
```